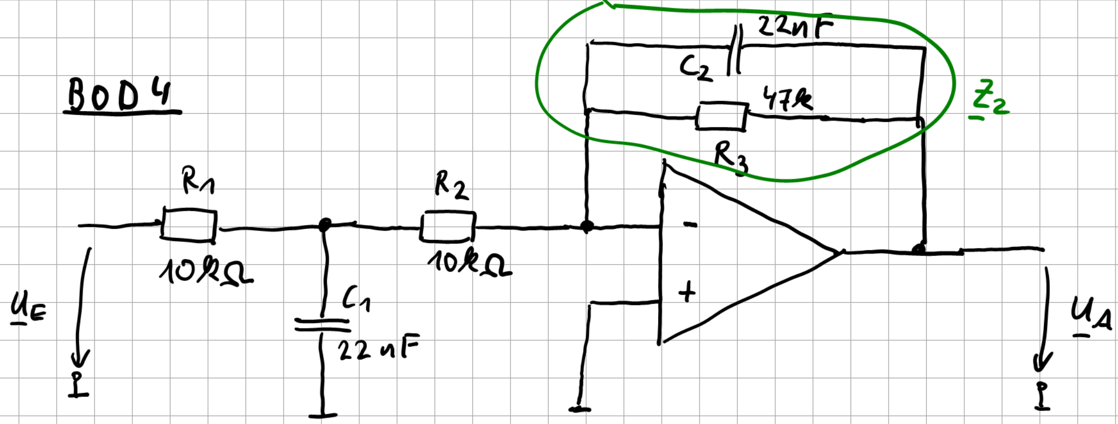
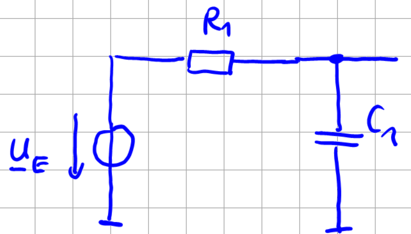
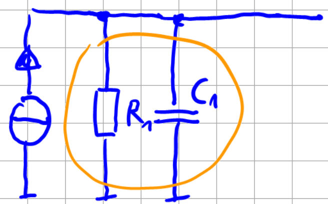
BOD 4

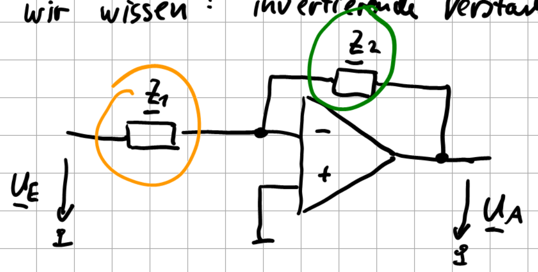




$\underline{I}_E = \frac{\underline{U}_E}{R_1}$

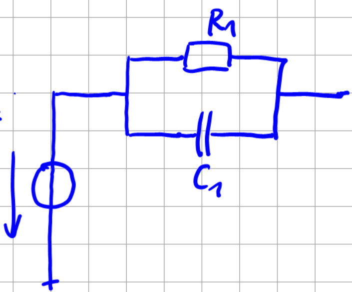


wir wissen: invertierende Verstärker



$$\frac{\underline{U}_A}{\underline{U}_E} = -\frac{\underline{Z}_2}{\underline{Z}_1}$$

$\underline{U}_{Ers} =$

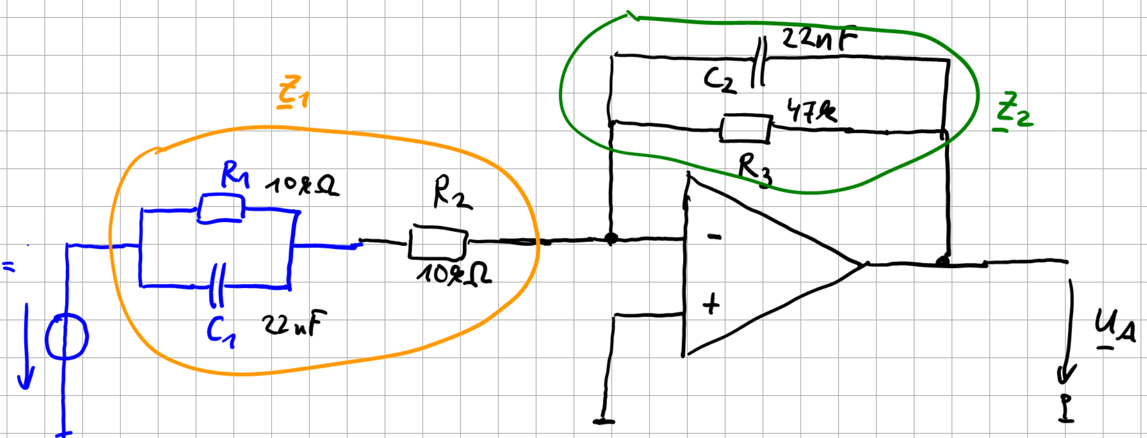


$$\underline{U}_{Ers} = \underline{I}_E \cdot (R_1 \parallel \underline{Z}_{C1})$$

$$= \underline{I}_E \cdot \frac{R_1 \cdot \frac{1}{j\omega C_1}}{R_1 + \frac{1}{j\omega C_1}} \cdot \frac{j\omega C_1}{j\omega C_1} = \underline{I}_E \cdot \frac{R_1}{1 + j\omega C_1 R_1}$$

$$\underline{U}_{Ers} = \frac{\underline{U}_E}{\cancel{R_1}} \cdot \frac{\cancel{R_1}}{1 + j\omega R_1 C_1} = \underline{U}_E \cdot \frac{1}{1 + j\omega R_1 C_1}$$

$\underline{U}_{Ers} =$



EIT

Amplituden- und Phasengang (Bode-Diagramm)

Aufgabe:

$$\underline{A}_{ges} = \boxed{K} \cdot \boxed{\frac{1}{1+j\frac{f}{f_1}}} \cdot \boxed{\frac{1}{1+j\frac{f}{f_2}}}$$

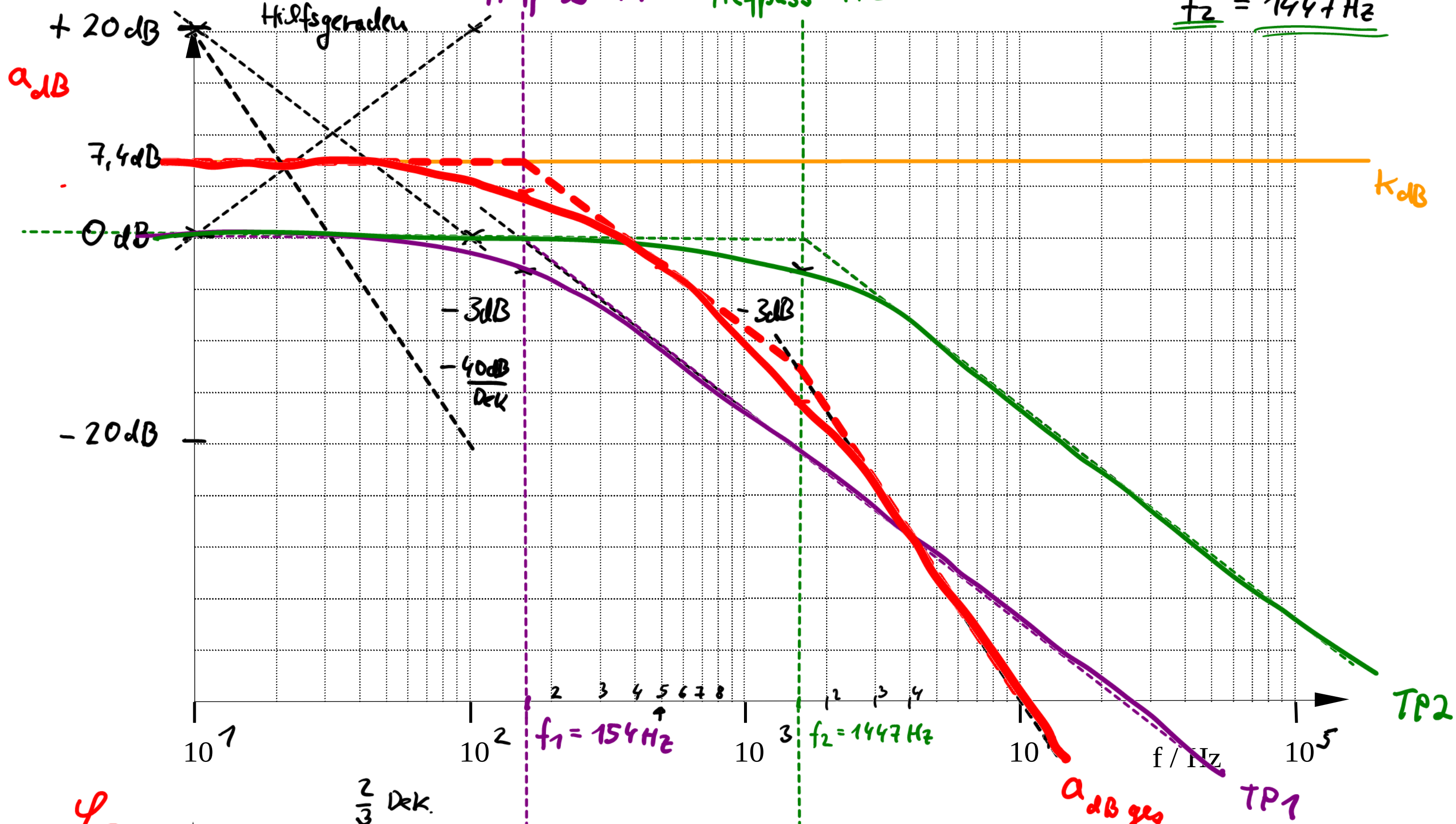
Tiefpass TP1   Tiefpass TP2

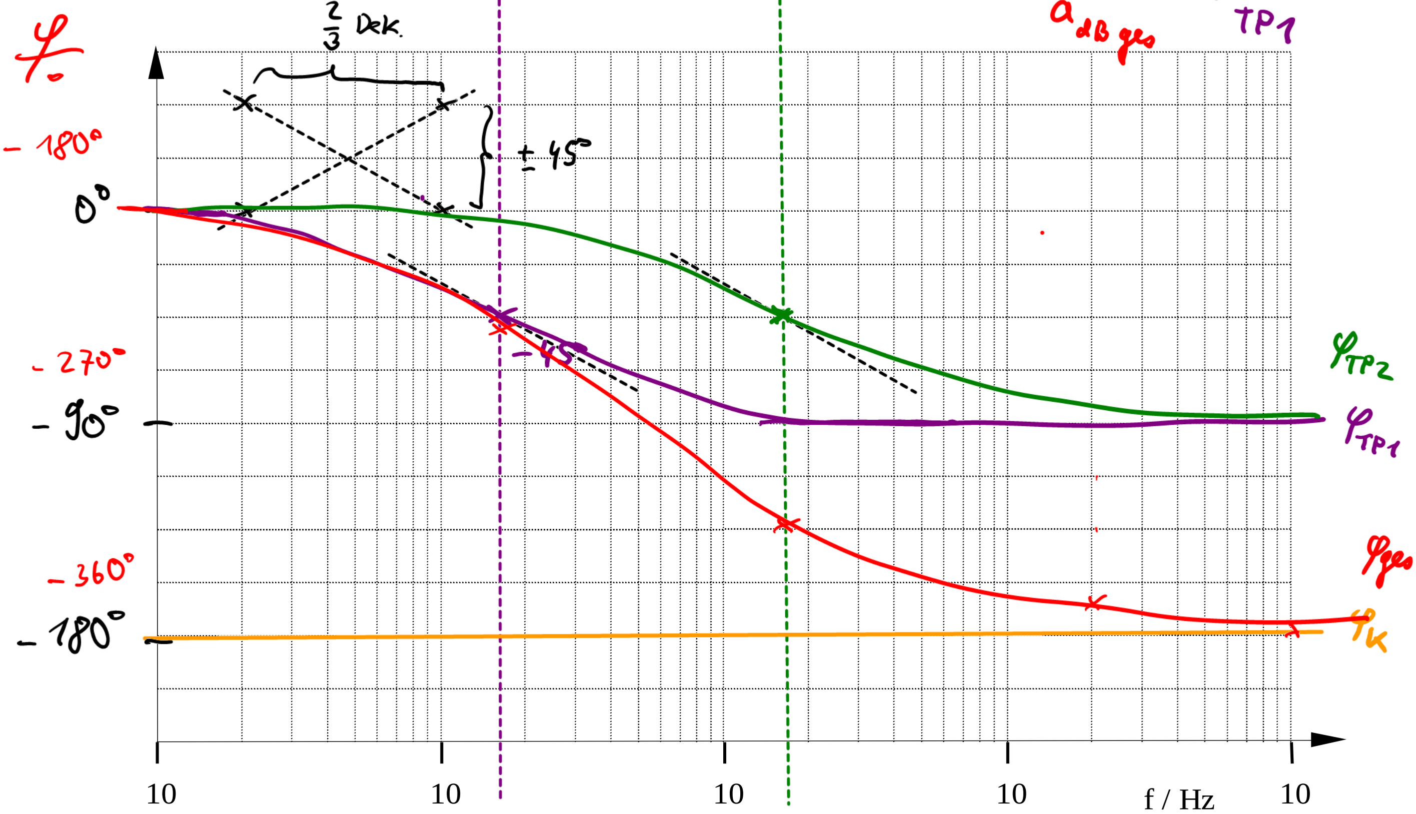
$K = -2{,}34$

$K_{dB} = 7{,}4\,dB$

$f_1 = 154\,Hz$

$f_2 = 1447\,Hz$





$$\underline{Z}_2 = (\underline{Z}_{C2} \parallel R_3) = \frac{\frac{1}{j\omega C_2} \cdot R_3}{\frac{1}{j\omega C_2} + R_3} = \frac{R_3}{1 + j\omega C_2 R_3}$$

$$\underline{Z}_1 = (R_1 \parallel \underline{Z}_{C1}) + R_2 = \frac{\frac{1}{j\omega C_1} \cdot R_1}{\frac{1}{j\omega C_1} + R_1} + R_2$$

$$= \frac{R_1}{1 + j\omega C_1 R_1} + R_2$$

$$= \frac{R_1 + R_2 \cdot (1 + j\omega C_1 R_1)}{1 + j\omega C_1 R_1}$$

$$= \frac{R_1 + R_2 + j\omega C_1 R_1 R_2}{1 + j\omega C_1 R_1}$$

$$\frac{\underline{U}_A}{\underline{U}_{Ers}} = -\frac{\underline{Z}_2}{\underline{Z}_1} = -\frac{\frac{R_3}{1 + j\omega C_2 R_3}}{\frac{R_1 + R_2 + j\omega C_1 R_1 R_2}{1 + j\omega C_1 R_1}}$$

$$= -\frac{R_3}{(1 + j\omega C_2 R_3)} \cdot \frac{(1 + j\omega C_1 R_1)}{(R_1 + R_2 + j\omega C_1 R_1 R_2)}$$

$$\frac{\underline{U}_A}{\underline{U}_E \cdot \frac{1}{1 + j\omega R_1 C_1}} = -\frac{R_3}{(1 + j\omega C_2 R_3)} \cdot \frac{\cancel{(1 + j\omega C_1 R_1)}}{(R_1 + R_2 + j\omega C_1 R_1 R_2)} \cdot \cancel{\frac{1}{1 + j\omega R_1 C_1}}$$

$$\frac{\underline{U}_A}{\underline{U}_E} = -\frac{R_3}{1 + j\omega \underbrace{C_2 R_3}_{\tau_1}} \cdot \frac{1}{(R_1 + R_2) + j\omega C_1 R_1 R_2}$$

b) $$\underline{A}_{ges}(j\omega) = K \cdot \frac{1}{\left(1 + j\frac{\omega}{\omega_1}\right) \cdot \left(1 + j\frac{\omega}{\omega_2}\right)}$$ Vorgabe

$$\frac{\underline{U}_A}{\underline{U}_E} = -R_3 \; \frac{1}{1 + j\omega C_2 R_3} \; \frac{1}{R_1 + R_2} \; \frac{1}{1 + j\omega \underbrace{C_1 \frac{R_1 R_2}{R_1 + R_2}}_{\tau_2}}$$

$$= -\frac{R_3}{R_1 + R_2} \cdot \frac{1}{1 + j\omega \underbrace{C_2 R_3}_{\tau_1}} \cdot \frac{1}{1 + j\omega \underbrace{C_1 \frac{R_1 R_2}{R_1 + R_2}}_{\tau_2}}$$

Zahlenwerte:

$$K = -\frac{R_3}{R_1+R_2} = -\frac{47k}{10k+10k} = -2{,}35$$

$$\underline{K_{dB}} = 20 \cdot \log_{10} |-2{,}35| = \underline{7{,}4\ dB}$$

$$\tau_1 = C_2 R_3 = 22nF \cdot 47k\Omega = 1{,}034\ ms$$

$$\omega_1 = \frac{1}{\tau_1} = 967\ \frac{1}{s}$$

$$\underline{f_1} = \frac{\omega_1}{2\pi} = \underline{\underline{154\ Hz}}$$

$$\tau_2 = C_1 \cdot \frac{R_1 R_2}{R_1+R_2} = 22nF \cdot \frac{10k\Omega \cdot 10k\Omega}{10k\Omega + 10k\Omega} = 110\mu s$$

$$\omega_2 = \frac{1}{\tau_2} = 9091\ \frac{1}{s}$$

$$\underline{\underline{f_2}} = \frac{\omega_2}{2\pi} = \underline{\underline{1447\ Hz}}$$

$$\frac{\underline{U_A}}{\underline{U_E}} = \underbrace{-\frac{R_3}{R_1+R_2}}_{K} \cdot \underbrace{\frac{1}{1+j\frac{f}{f_1}}}_{TP1} \cdot \underbrace{\frac{1}{1+j\frac{f}{f_2}}}_{TP2}$$

c) Grenzwerte bestimmen für $\omega \to 0$, $\omega \to \infty$

$$\lim_{\omega \to 0} \underline{A}(j\omega) = \lim_{\substack{\omega \to 0 \\ f \to 0}} \left(-\frac{R_3}{R_1+R_2} \cdot \frac{1}{1+\cancel{j\frac{f}{f_1}}} \cdot \frac{1}{1+\cancel{j\frac{f}{f_2}}} \right)$$

Vorgehen $\measuredangle$ ≙ „$1+j\frac{f}{f_1}$"

$$= \ominus \frac{R_3}{R_1+R_2} \qquad \varphi = \pm 180°$$

$$\varphi(\omega \to 0) = \pm 180°$$

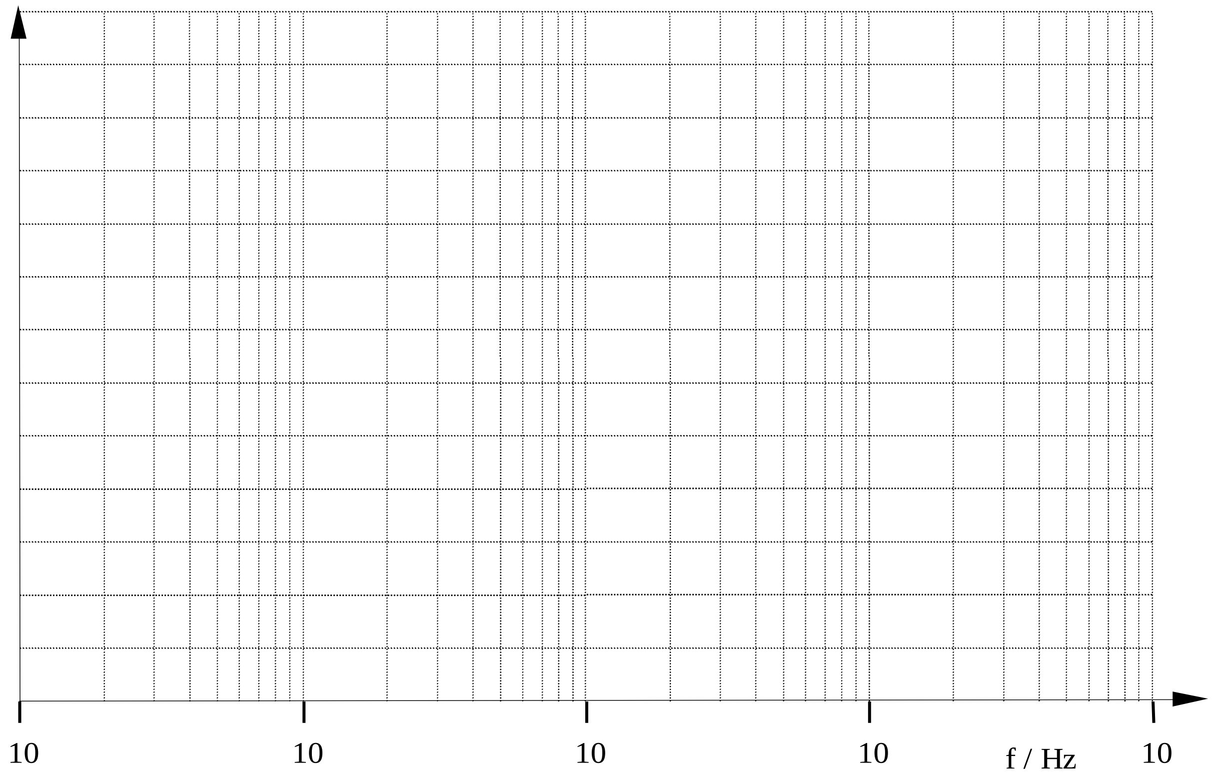
$$\lim_{\substack{\omega \to \infty \\ f \to \infty}} \underline{A}(j\omega) = \lim_{f \to \infty} \left(\cancel{-} \frac{R_3}{R_1 + R_2} \cdot \frac{1}{\cancel{1+} \cancel{j} \frac{f}{f_1}} \cdot \frac{1}{\cancel{1+} \cancel{j} \frac{f}{f_2}} \right)$$

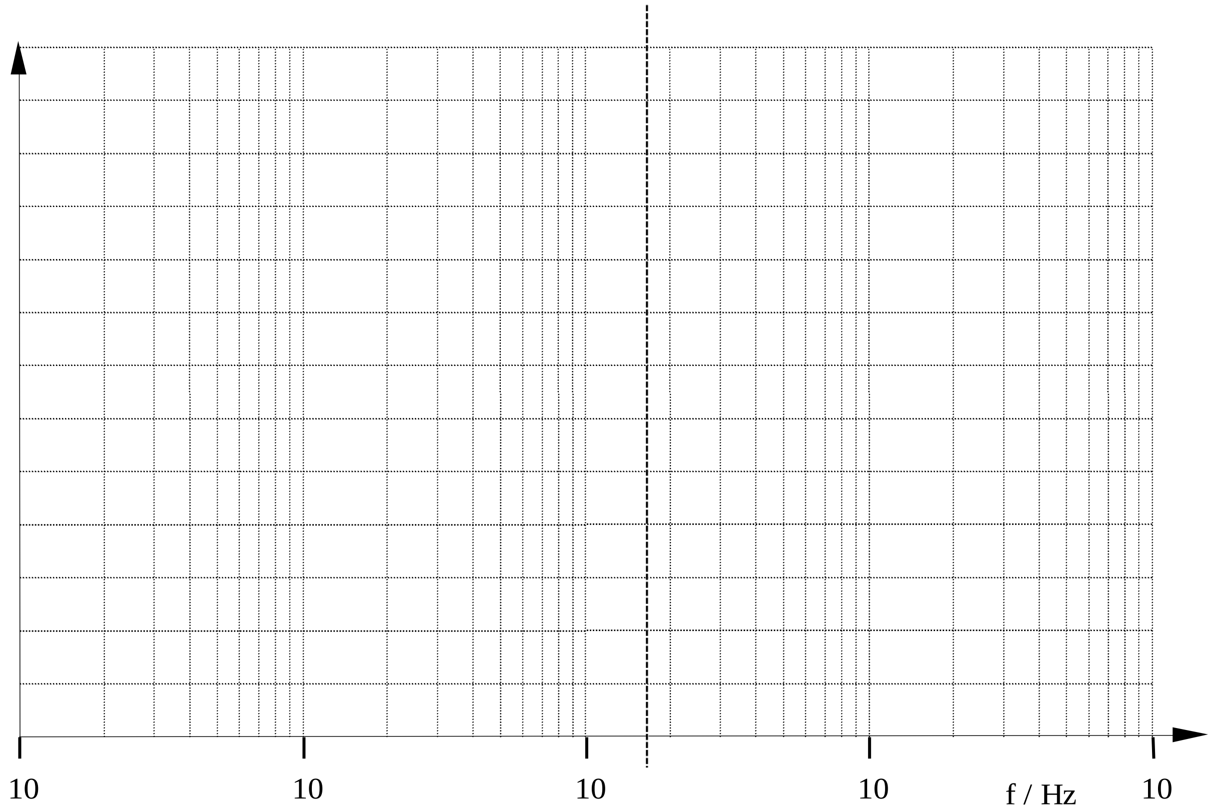
$$= \lim_{f \to \infty} \left(\frac{R_3}{R_1 + R_2} \quad \frac{f_1 \cdot f_2}{f^2} \right)$$

$$= 0.0$$

$$\varphi(\omega \to \infty) = 0^\circ$$

Aufgabe:





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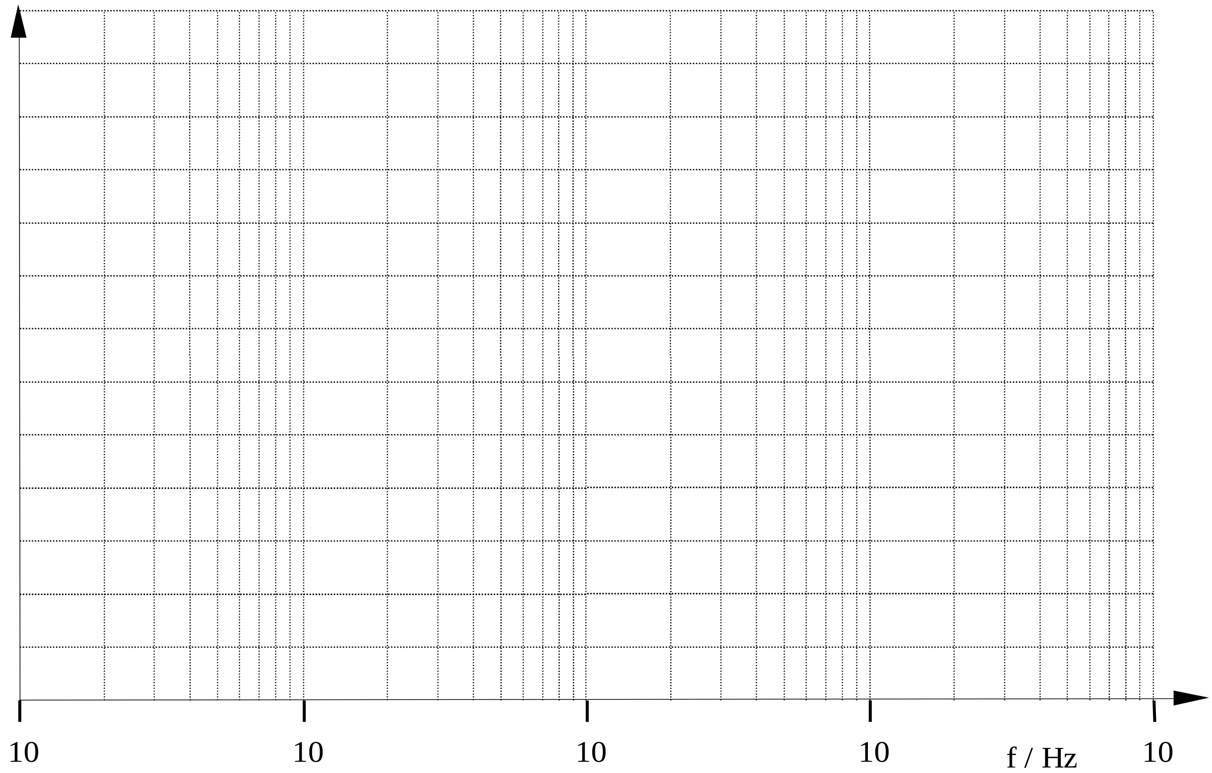

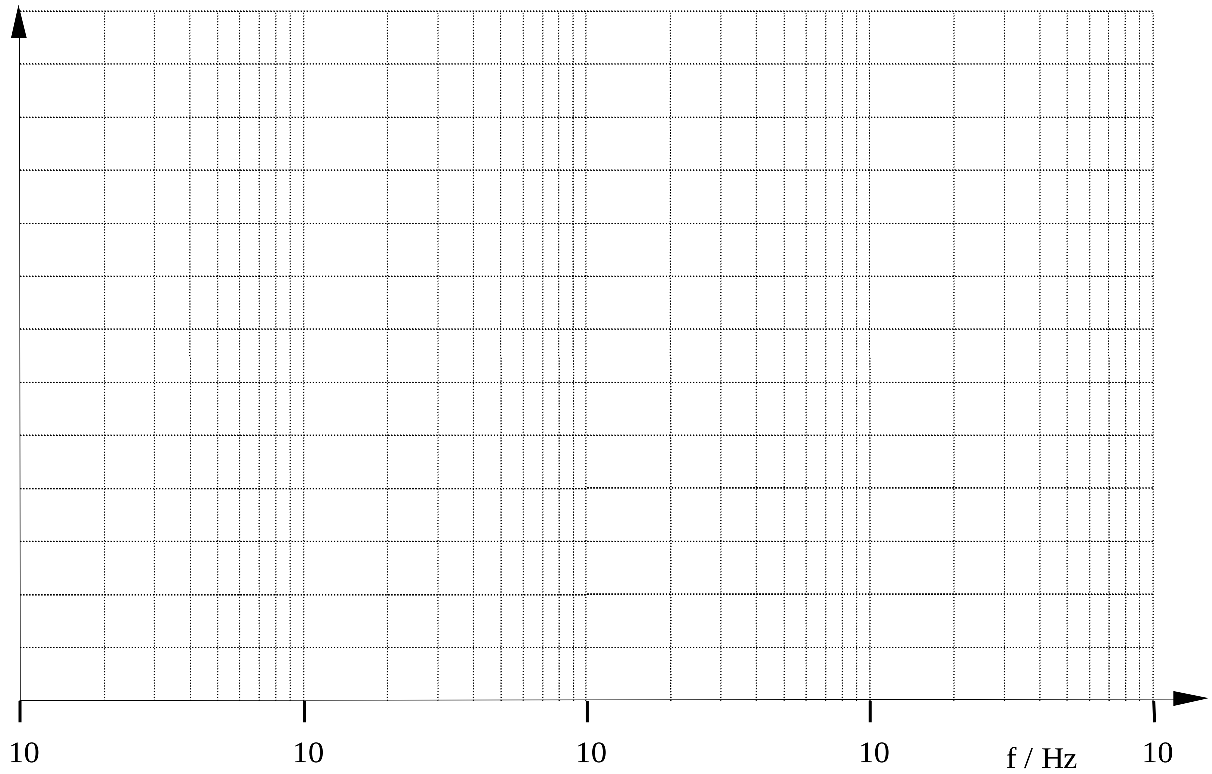

Aufgabe: